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BCSE355L - Cloud Architecture Design

Project Title -: Reinforcement Learning-Based Intelligent Queue Optimization Framework for Railway Reservation Systems

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ABSTRACT

The rapid growth of railway passengers has significantly increased the complexity of reservation management, resulting in long waiting lists, inefficient seat allocation, and delayed confirmation of tickets. Existing railway reservation systems primarily rely on static First-Come-First-Serve (FCFS) mechanisms that do not adapt to changing passenger demand or historical reservation patterns. Consequently, available seats are not always utilized efficiently, and passengers often experience long waiting periods despite frequent ticket cancellations.

This project proposes a **Reinforcement Learning-Based Intelligent Queue Optimization Framework for Railway Reservation Systems**, designed to improve waiting-list management through adaptive decision-making. Unlike conventional reservation systems, the proposed framework employs a Reinforcement Learning (RL) agent that continuously learns from historical reservation data, passenger behavior, cancellation patterns, and seat occupancy to determine optimal queue reallocation strategies.

The framework aims to minimize passenger waiting time, maximize seat utilization, and ensure fair and efficient ticket allocation.

The proposed system is deployed on Amazon Web Services (AWS), enabling scalability, reliability, and real-time processing. Amazon EC2 hosts the application, Amazon RDS stores reservation records, Amazon S3 manages datasets and logs, AWS Lambda automates queue processing, Amazon SNS provides passenger notifications, Amazon Cognito manages authentication, and Amazon CloudWatch monitors system performance.

By integrating intelligent learning with cloud computing, the proposed framework provides a scalable and adaptive reservation management solution capable of handling high passenger demand while improving operational efficiency and user satisfaction.

PROBLEM STATEMENT

Modern railway reservation systems handle millions of ticket requests every day. During peak travel periods, a large number of passengers are placed on waiting lists due to limited seat availability. Current reservation systems generally allocate cancelled seats using predefined First-Come-First-Serve (FCFS) rules without considering historical demand, passenger travel patterns, or cancellation probabilities. As a result, seats may remain underutilized, waiting times increase, and reservation management becomes less efficient.

Additionally, high booking traffic during Tatkal and festival seasons places significant computational load on reservation servers, affecting response time and user experience. Existing systems lack intelligent decision-making capabilities that can dynamically optimize seat allocation and improve overall reservation efficiency.

Therefore, there is a need for an intelligent, scalable, and adaptive queue optimization framework capable of learning from historical reservation behavior and making optimal seat allocation decisions in real time.

EXISTING CHALLENGES

- Current railway reservation systems primarily follow a **First-Come-First-Serve (FCFS)** approach, which cannot adapt to changing passenger demand or cancellation patterns.
- During peak booking periods (such as festivals and Tatkal bookings), the system experiences **high server load**, leading to slower response times.
- **Waiting-list management is static**, resulting in longer confirmation times and inefficient seat allocation.
- Cancelled tickets are reallocated using predefined rules instead of intelligent decision-making, which can reduce seat utilization.
- Passengers receive **limited real-time information** regarding queue movement and confirmation probability.
- Existing systems do not effectively use **historical booking and cancellation data** to improve future allocation decisions.

- Scaling the reservation infrastructure to handle millions of concurrent users is challenging without cloud-based resources

PROPOSED SOLUTION

The proposed system introduces a Reinforcement Learning-based queue optimization framework integrated with cloud computing services. Instead of relying solely on static FCFS rules, the framework continuously learns from reservation history, cancellation events, passenger demand, and seat occupancy to determine the most effective waiting-list allocation strategy.

The application is deployed on AWS, where cloud services provide scalable computing, secure data storage, real-time notification, and continuous monitoring. Whenever a ticket is cancelled, the RL agent evaluates the current queue state, selects the optimal passenger for seat allocation based on the learned policy, updates the reservation database, and immediately notifies the passenger.

This intelligent approach aims to reduce waiting times, improve seat utilization, increase system efficiency, and enhance passenger satisfaction.

AWS SERVICES USED

1. Amazon EC2
2. Amazon S3
3. Amazon RDS
4. AWS Lambda
5. Amazon API Gateway
6. Amazon Cognito
7. Amazon SNS
8. Amazon CloudWatch

EXPECTED OUTCOMES

The proposed framework is expected to:

- Reduce passenger waiting time.
- Improve seat utilization after cancellations.
- Enhance fairness in waiting-list management.

- Provide scalable cloud-based reservation processing.
- Enable intelligent decision-making using Reinforcement Learning.
- Improve passenger satisfaction through timely notifications.
- Support future expansion to national railway reservation systems.